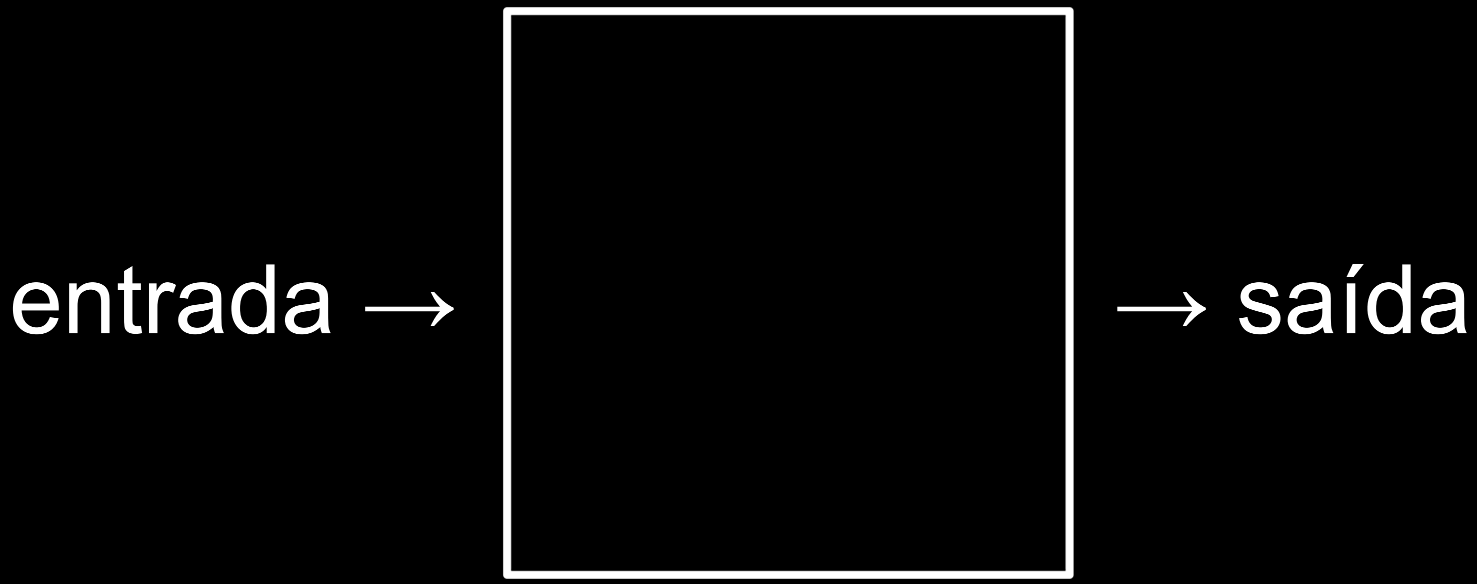
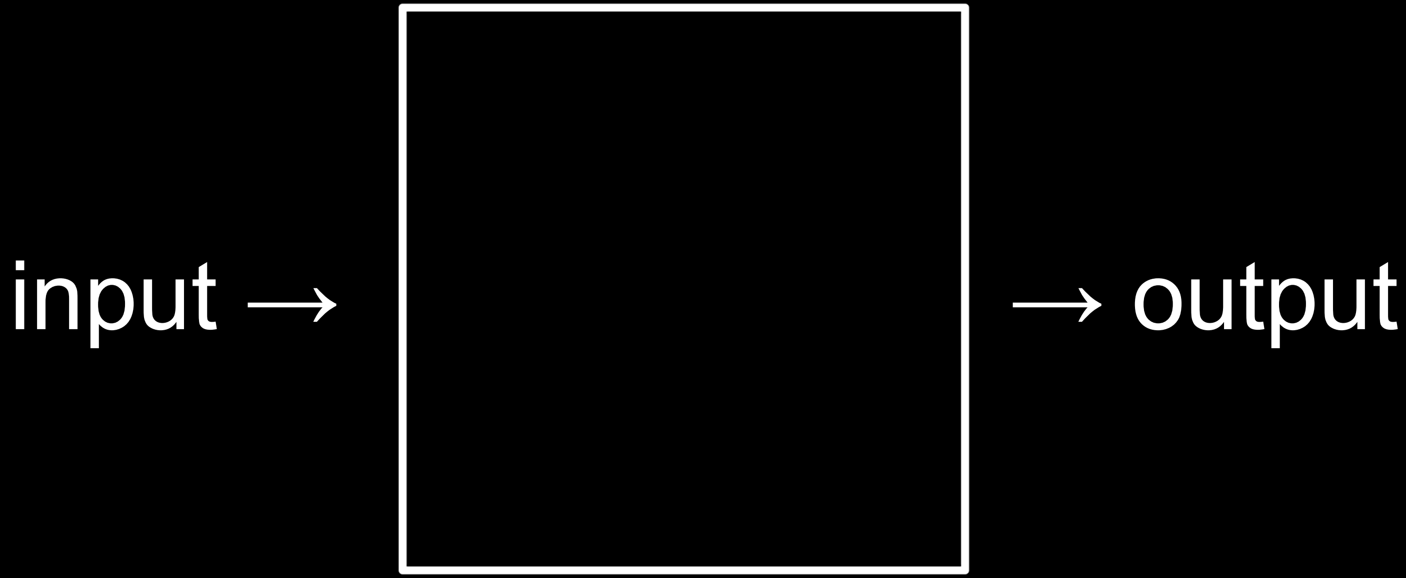
Lógica

ciência da computação

pensamento computacional

resolução de problemas


entrada →
→ saída


input →
→ output

unário

base 1

base 2

binário

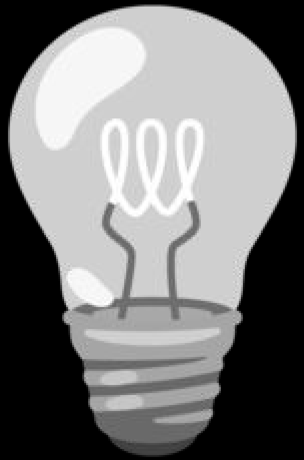
dígito binário

binary digit

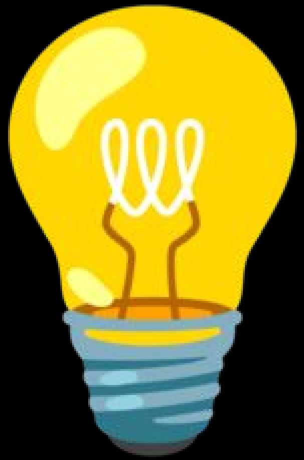
bi t

bit

0



1



base 10

decimal

123

1

123

10 1

123

100 10 1

123

100 10 1

123

100×1

100 10 1

123

$100 \times 1 \quad + \quad 10 \times 2$

100 10 1

123

$$100 \times 1 \quad + \quad 10 \times 2 \quad + \quad 1 \times 3$$

100 10 1

123

100 + 20 + 3

123

100	10	1
#	#	#

10^2 10^1 10^0

# # #

10^2 10^1 10^0

\# \# \#

100	10	1
#	#	#

4 2 1

000

4 2 1

001

4	2	1
0	1	0

4 2 1

011

4 2 1

100

4	2	1
1	0	1

4 2 1

110

4 2 1

111

4	2	1
0	0	0

8 4 2 1

1000

byte

128	64	32	16	8	4	2	1
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128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	1

A

128	64	32	16	8	4	2	1
0	1	0	0	0	0	0	1

65

ASCII

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3	ETX	19	DC3	35	#	51	3	67	C	83	S	99	c	115	s
4	EOT	20	DC4	36	$	52	4	68	D	84	T	100	d	116	t
5	ENQ	21	NAK	37	%	53	5	69	E	85	U	101	e	117	u
6	ACK	22	SYN	38	&	54	6	70	F	86	V	102	f	118	v
7	BEL	23	ETB	39	'	55	7	71	G	87	W	103	g	119	w
8	BS	24	CAN	40	(	56	8	72	H	88	X	104	h	120	x
9	HT	25	EM	41	)	57	9	73	I	89	Y	105	i	121	y
10	LF	26	SUB	42	*	58	:	74	J	90	Z	106	j	122	z
11	VT	27	ESC	43	+	59	;	75	K	91	[	107	k	123	{
12	FF	28	FS	44	,	60	<	76	L	92	\	108	l	124	|
13	CR	29	GS	45	-	61	=	77	M	93	]	109	m	125	}
14	SO	30	RS	46	.	62	>	78	N	94	^	110	n	126	~
15	SI	31	US	47	/	63	?	79	O	95	_	111	o	127	DEL

0	NUL	16	DLE	32	SP	48	0	64	@	80	P	96	`	112	p
1	SOH	17	DC1	33	!	49	1	65	A	81	Q	97	a	113	q
2	STX	18	DC2	34	"	50	2	66	B	82	R	98	b	114	r
3	ETX	19	DC3	35	#	51	3	67	C	83	S	99	c	115	s
4	EOT	20	DC4	36	$	52	4	68	D	84	T	100	d	116	t
5	ENQ	21	NAK	37	%	53	5	69	E	85	U	101	e	117	u
6	ACK	22	SYN	38	&	54	6	70	F	86	V	102	f	118	v
7	BEL	23	ETB	39	'	55	7	71	G	87	W	103	g	119	w
8	BS	24	CAN	40	(	56	8	72	H	88	X	104	h	120	x
9	HT	25	EM	41	)	57	9	73	I	89	Y	105	i	121	y
10	LF	26	SUB	42	*	58	:	74	J	90	Z	106	j	122	z
11	VT	27	ESC	43	+	59	;	75	K	91	[	107	k	123	{
12	FF	28	FS	44	,	60	<	76	L	92	\	108	l	124	|
13	CR	29	GS	45	-	61	=	77	M	93	]	109	m	125	}
14	SO	30	RS	46	.	62	>	78	N	94	^	110	n	126	~
15	SI	31	US	47	/	63	?	79	O	95	_	111	o	127	DEL

01001111 01001001 00100001

79 73 33

0	NUL	16	DLE	32	SP	48	0	64	@	80	P	96	`	112	p
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4	EOT	20	DC4	36	$	52	4	68	D	84	T	100	d	116	t
5	ENQ	21	NAK	37	%	53	5	69	E	85	U	101	e	117	u
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7	BEL	23	ETB	39	'	55	7	71	G	87	W	103	g	119	w
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9	HT	25	EM	41	)	57	9	73	I	89	Y	105	i	121	y
10	LF	26	SUB	42	*	58	:	74	J	90	Z	106	j	122	z
11	VT	27	ESC	43	+	59	;	75	K	91	[	107	k	123	{
12	FF	28	FS	44	,	60	<	76	L	92	\	108	l	124	|
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4	EOT	20	DC4	36	$	52	4	68	D	84	T	100	d	116	t
5	ENQ	21	NAK	37	%	53	5	69	E	85	U	101	e	117	u
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7	BEL	23	ETB	39	'	55	7	71	G	87	W	103	g	119	w
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11	VT	27	ESC	43	+	59	;	75	K	91	[	107	k	123	{
12	FF	28	FS	44	,	60	<	76	L	92	\	108	l	124	|
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14	SO	30	RS	46	.	62	>	78	N	94	^	110	n	126	~
15	SI	31	US	47	/	63	?	79	O	95	_	111	o	127	DEL

O	I	!
79	73	33

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5	ENQ	21	NAK	37	%	53	5	69	E	85	U	101	e	117	u
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10	LF	26	SUB	42	*	58	:	74	J	90	Z	106	j	122	z
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128	64	32	16	8	4	2	1
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0	NUL	16	DLE	32	SP	48	0	64	@	80	P	96	`	112	p
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14	SO	30	RS	46	.	62	>	78	N	94	^	110	n	126	~
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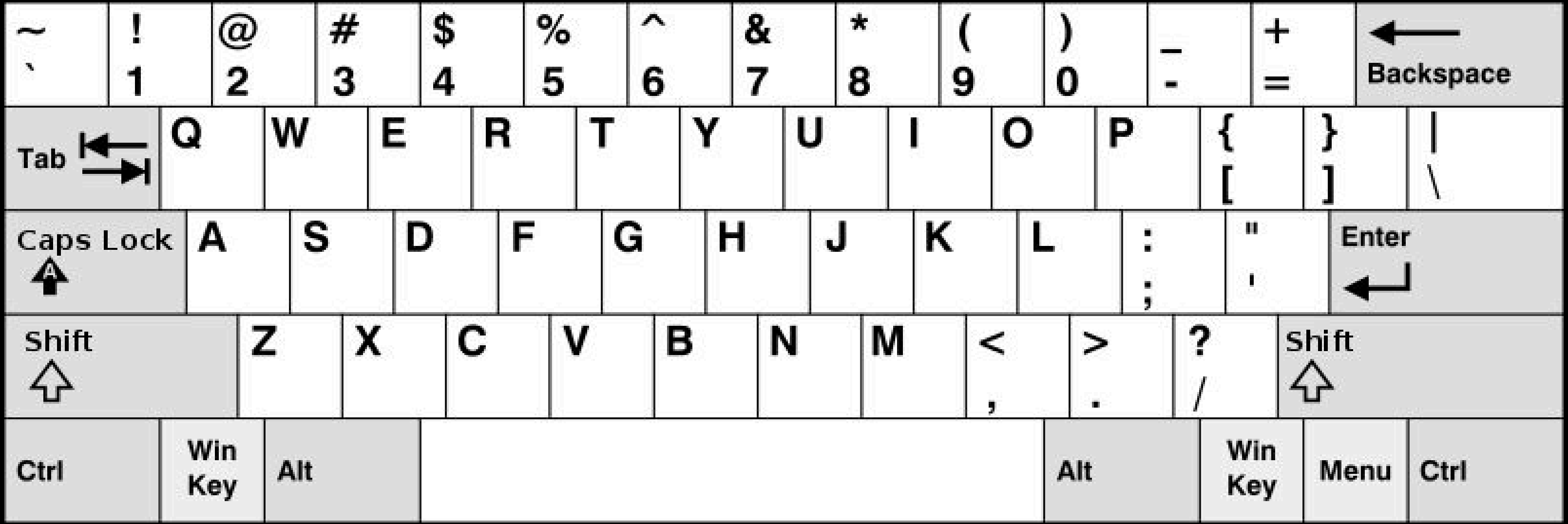
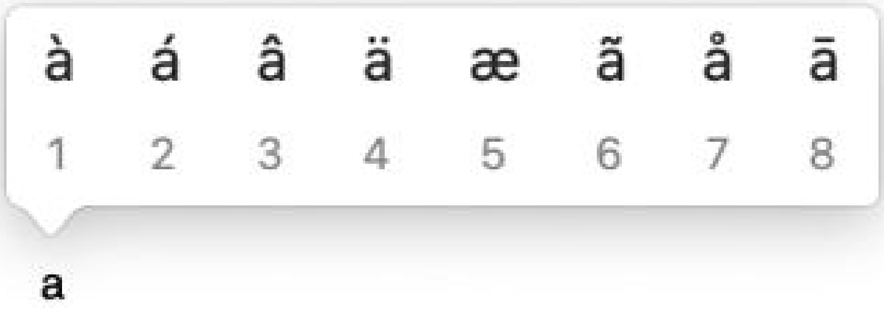
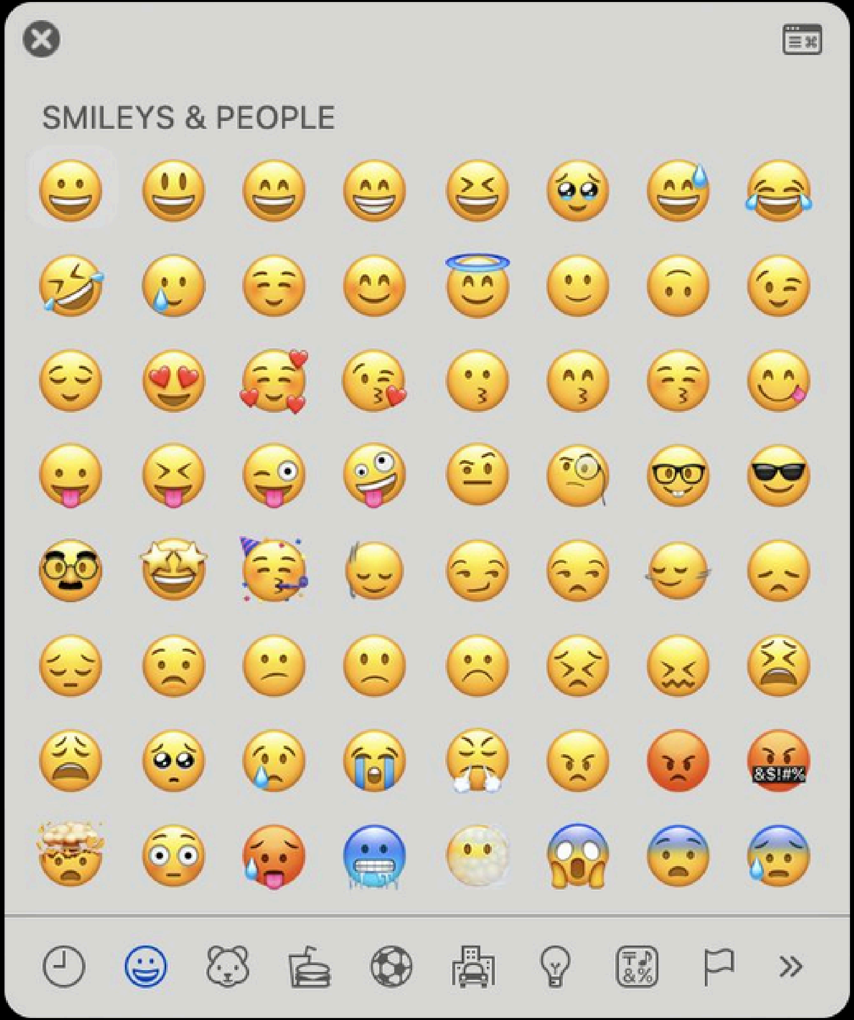

~ `
! 1
@ 2
# 3
$ 4
% 5
^ 6
& 7
* 8
(9
) 0
_ -
+ =
Backspace
Tab
Q W E R T Y U I O P
{ [
}]
| \
Caps Lock
A S D F G H J K L
: ;
" '
Enter
Shift
Z X C V B N M
< ,
> .
? /
Shift
Ctrl
Win Key
Alt
Alt
Win Key
Menu
Ctrl


à á â ä æ ã å ā
1 2 3 4 5 6 7 8
a


SMILEYS & PEOPLE

Unicode

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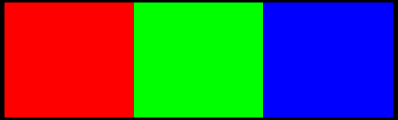
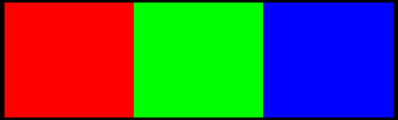
RGB

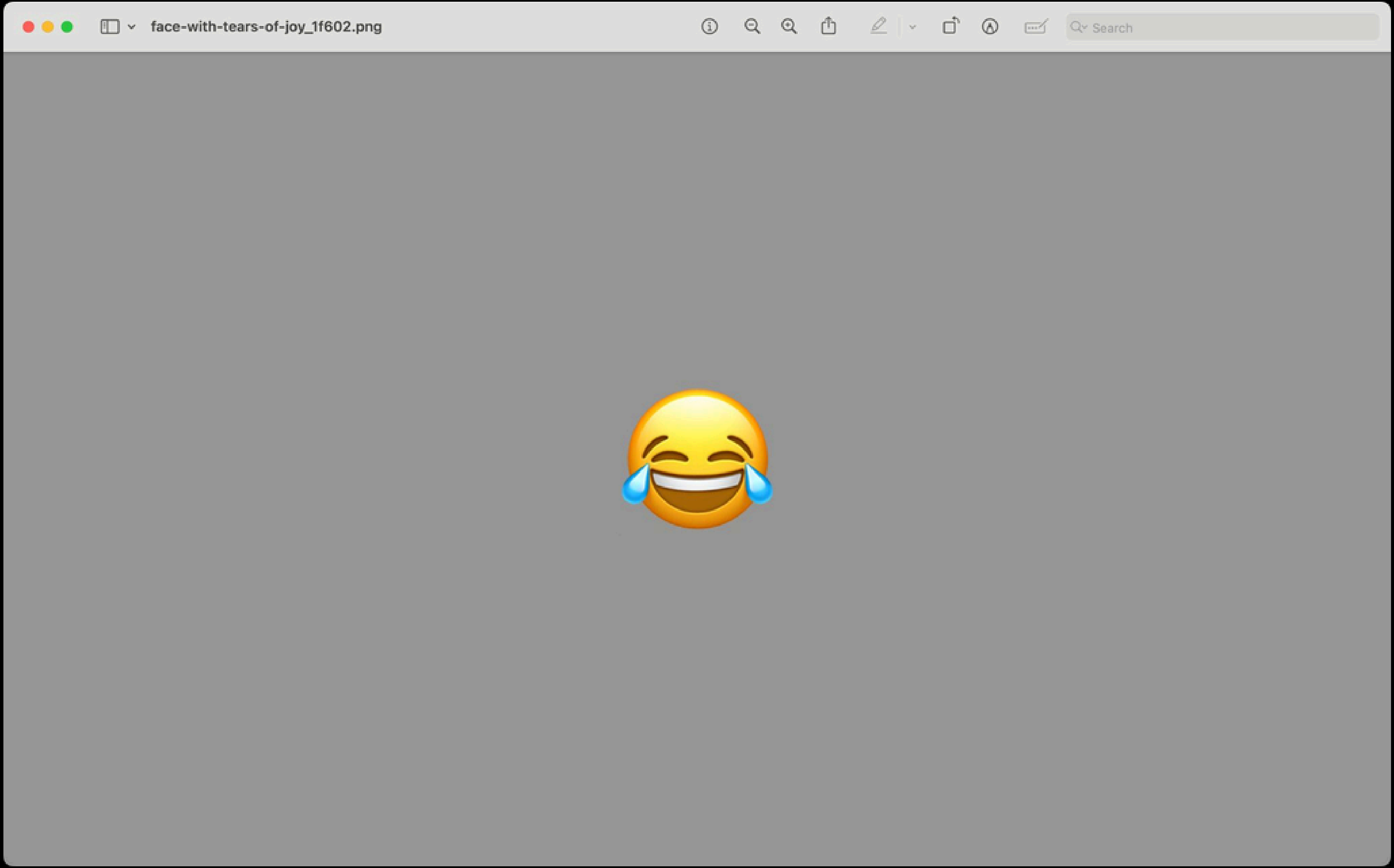

SONY



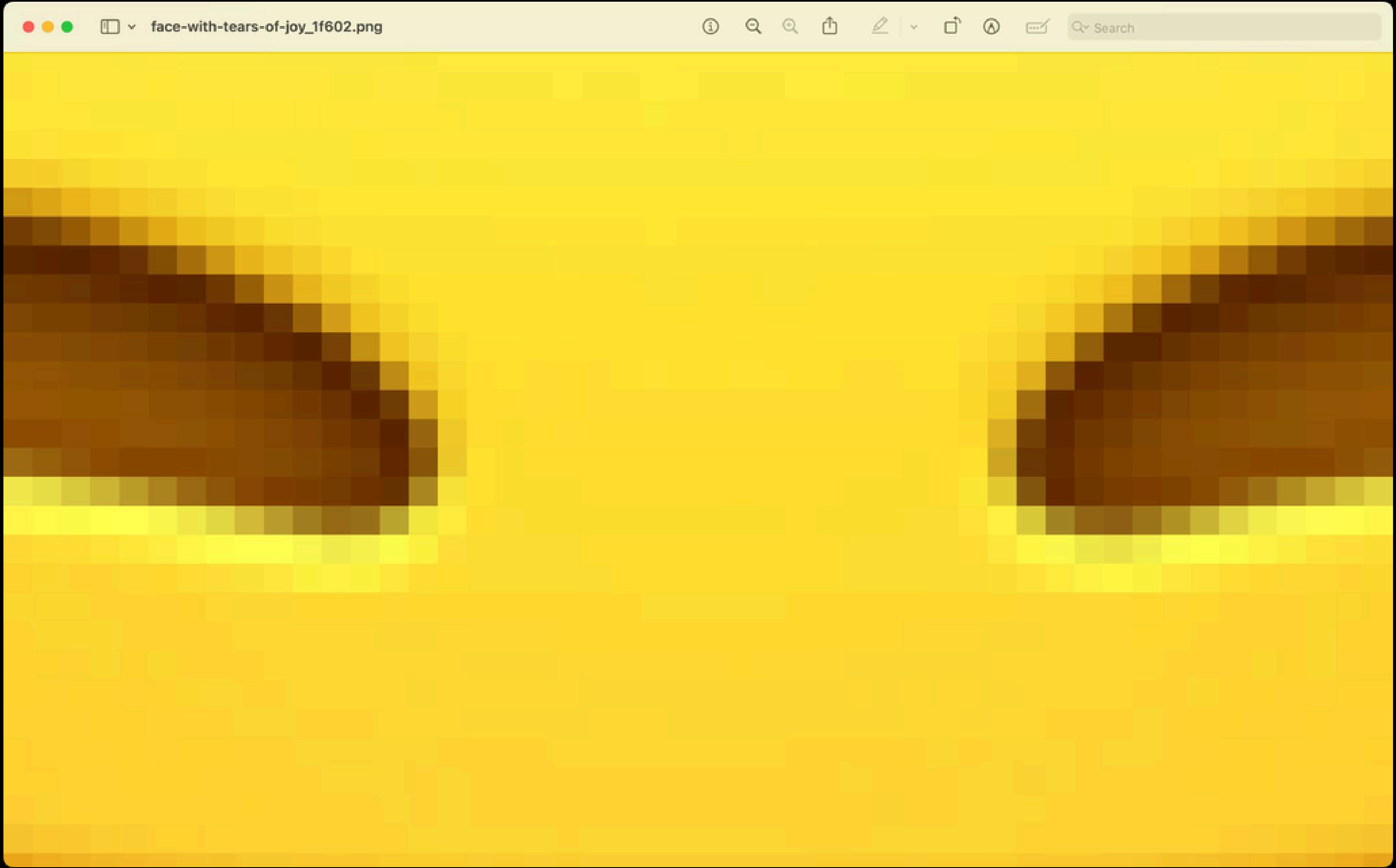
72 73 33

72 73 33




face-with-tears-of-joy_1f602.png
Search


face-with-tears-of-joy_1f602.png
Search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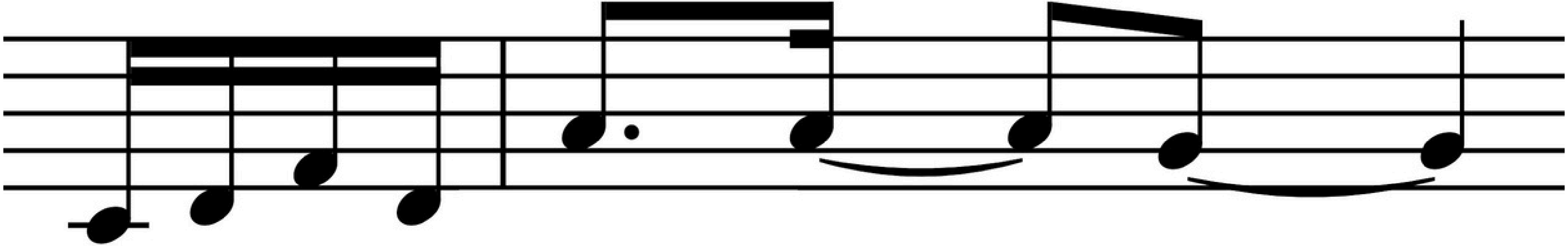
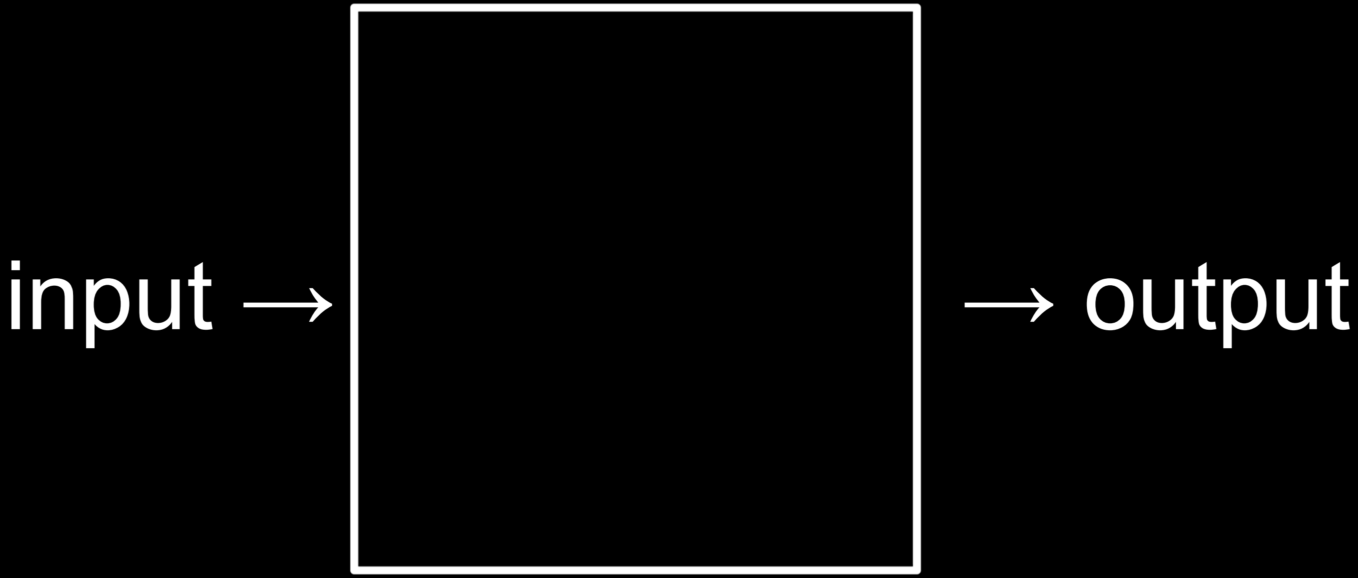
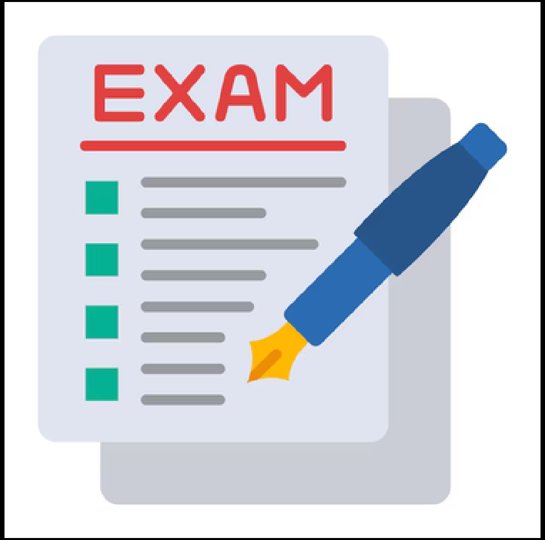
Search

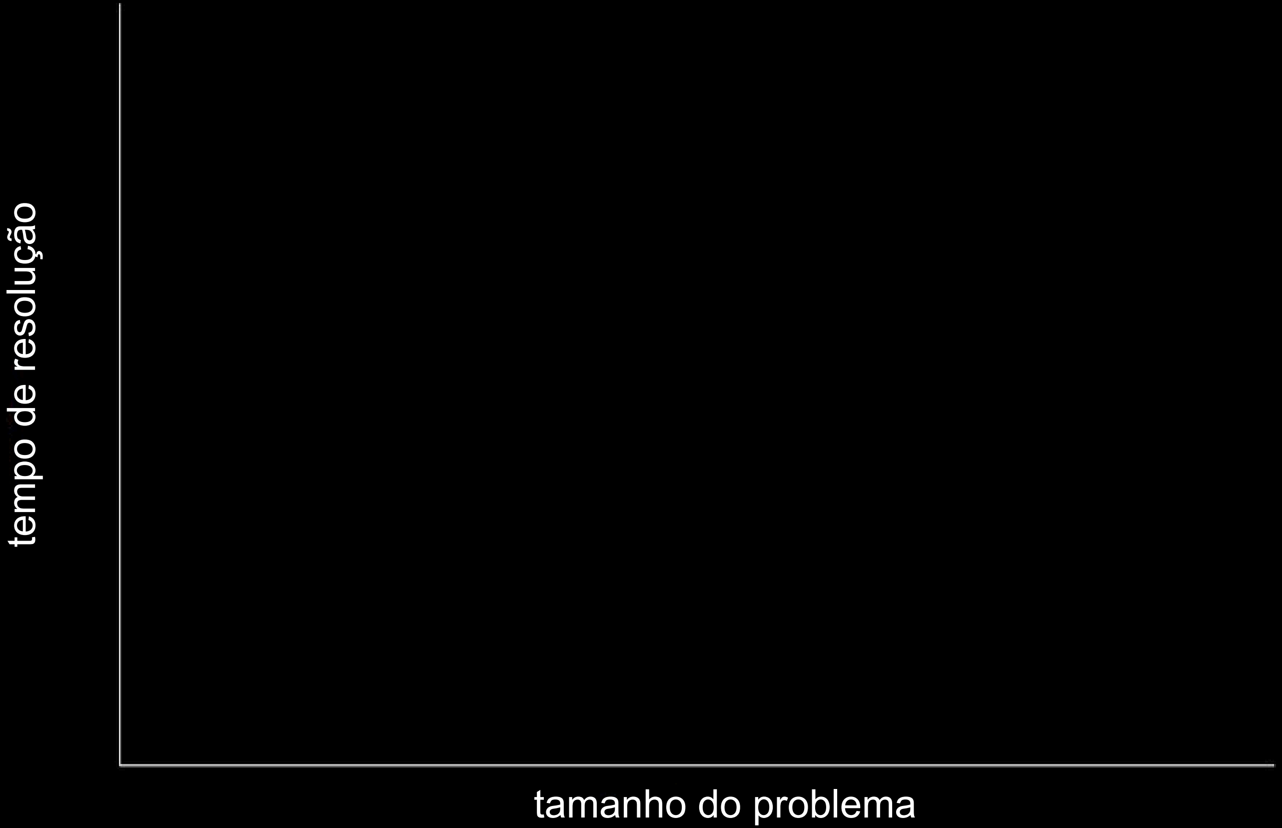



input →
→ output

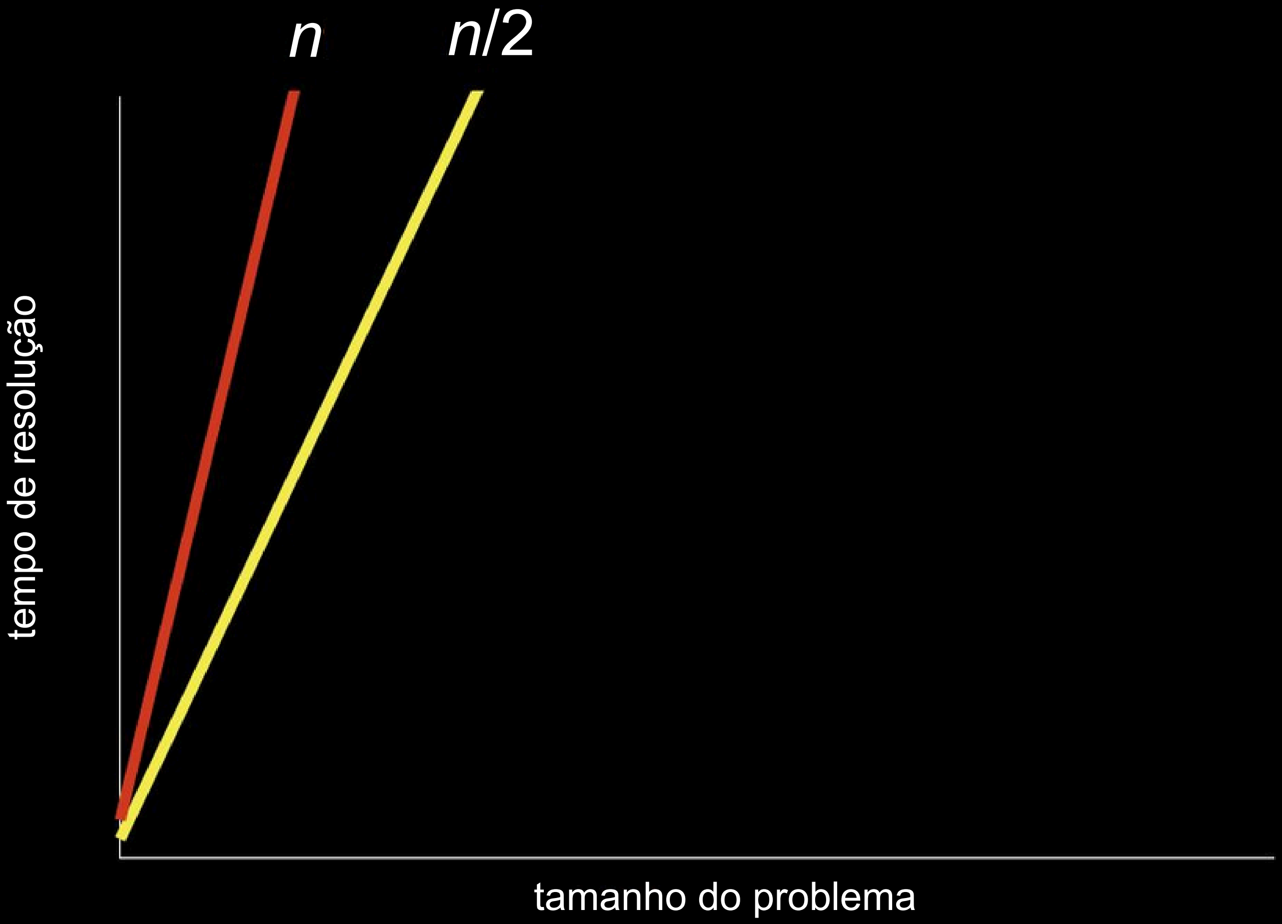
algoritmo

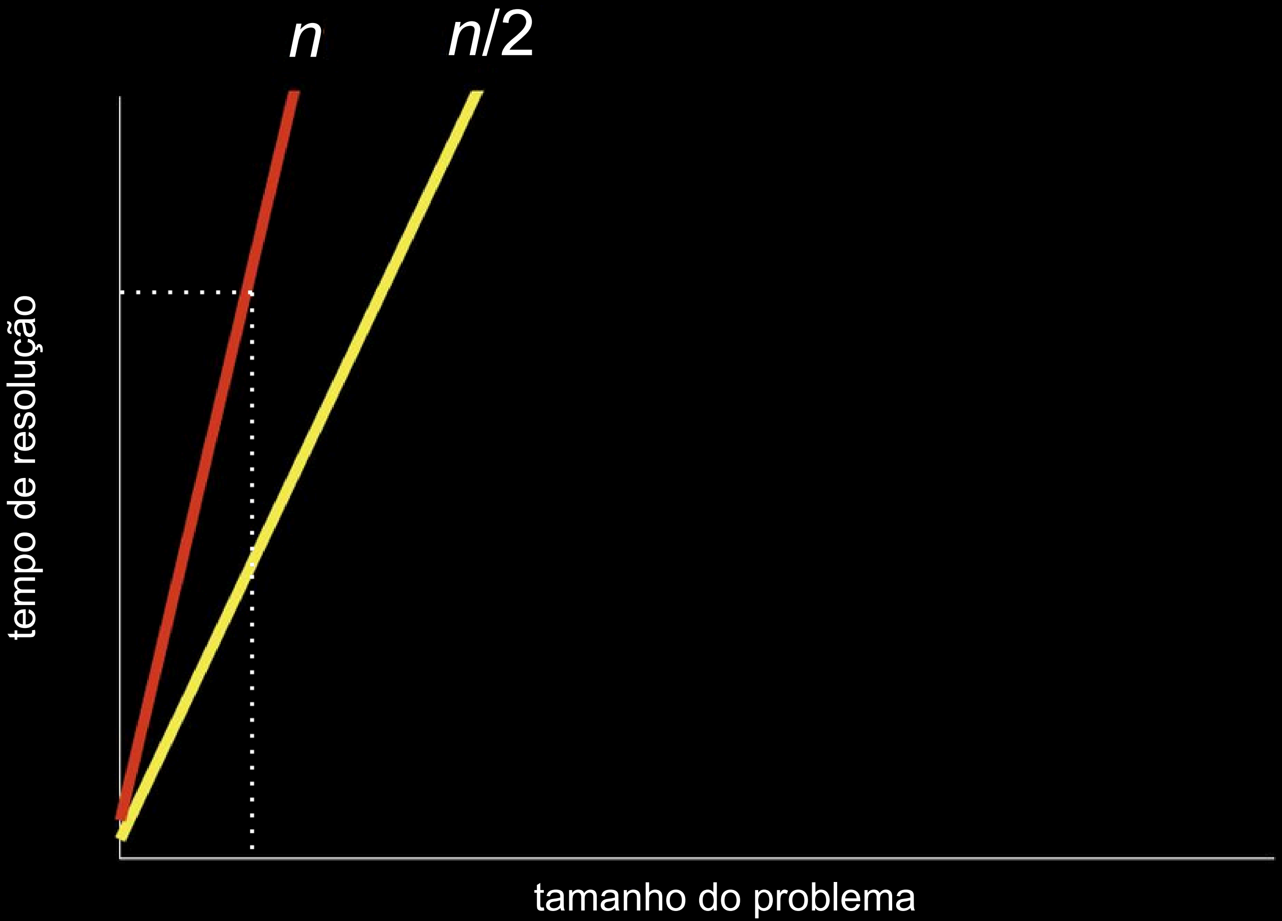
o computador é burro


EXAM

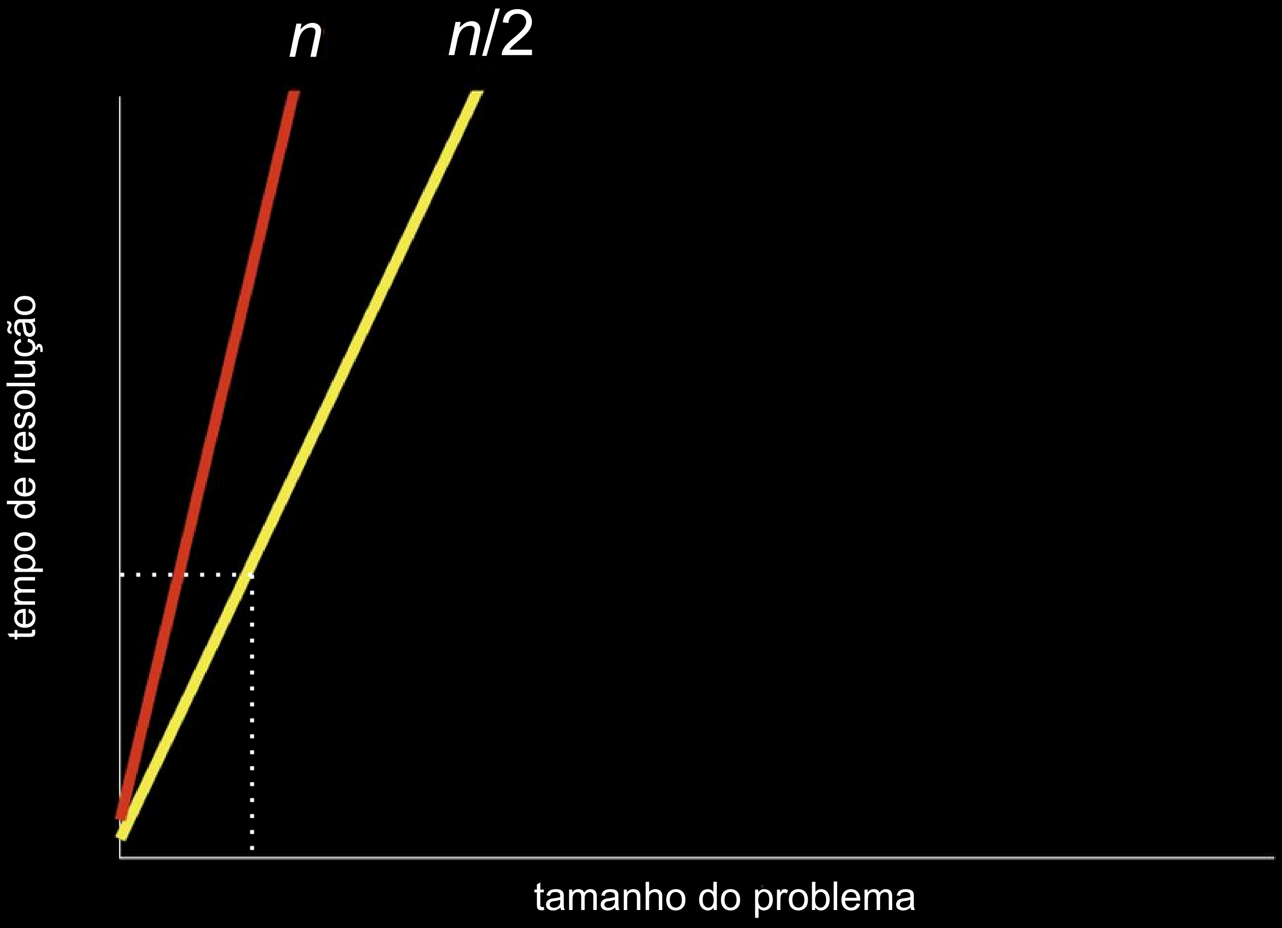

tempo de resolução
tamanho do problema

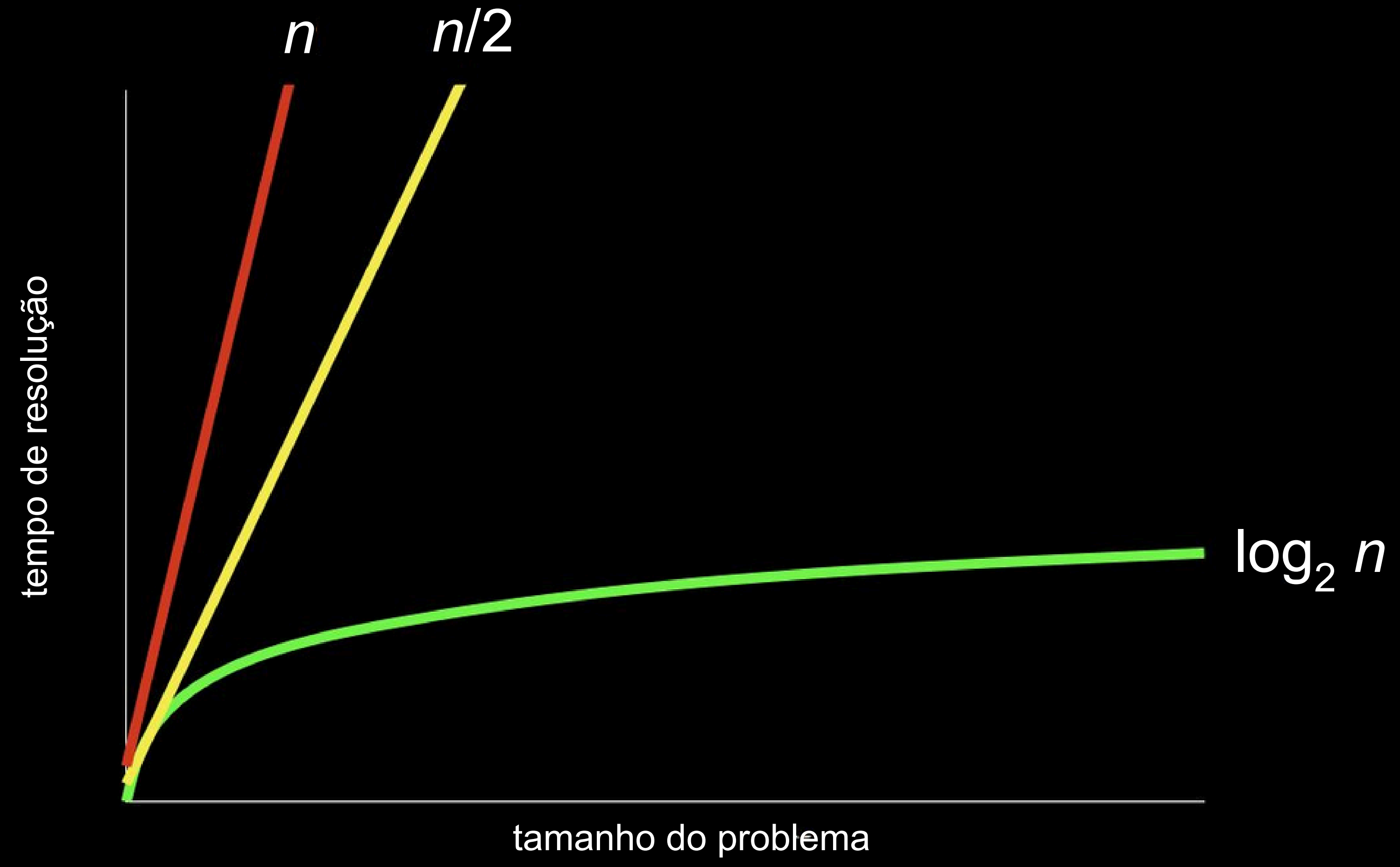

n
tempo de resolução
tamanho do problema


n
n/2
tempo de resolução
tamanho do problema



n
n/2
tempo de resolução
tamanho do problema


n
n/2
tempo de resolução
tamanho do problema



n
n/2
log2 n
tempo de resolução
tamanho do problema

algoritmo

código

pseudocódigo

```
1   Pegue o maço de provas
2   Abra no meio do maço de provas
3   Olhe o número do aluno
4   Se é o número procurado
5       Entregue a prova
6   Senão, se o número procurado é menor
7       Abra no meio da metade anterior do maço
8       Volte para a linha 3
9   Senão, se o número procurado é maior
10      Abra no meio da metade posterior do maço
11      Volte para a linha 3
12  Senão
13      Desista
```

```
1   Pegue o maço de provas
2   Abra no meio do maço de provas
3   Olhe o número do aluno
4   Se é o número procurado
5       Entregue a prova
6   Senão, se o número procurado é menor
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8       Volte para a linha 3
9   Senão, se o número procurado é maior
10      Abra no meio da metade posterior do maço
11      Volte para a linha 3
12  Senão
13      Desista
```

```
1   Pegue o maço de provas
2   Abra no meio do maço de provas
3   Olhe o número do aluno
4   Se é o número procurado
5       Entregue a prova
6   Senão, se o número procurado é menor
7       Abra no meio da metade anterior do maço
8       Volte para a linha 3
9   Senão, se o número procurado é maior
10      Abra no meio da metade posterior do maço
11      Volte para a linha 3
12  Senão
13      Desista
```

```
1   Pegue o maço de provas
2   Abra no meio do maço de provas
3   Olhe o número do aluno
4   Se é o número procurado
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9   Senão, se o número procurado é maior
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11      Volte para a linha 3
12  Senão
13      Desista
```

```
1   Pegue o maço de provas
2   Abra no meio do maço de provas
3   Olhe o número do aluno
4   Se é o número procurado
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9   Senão, se o número procurado é maior
10      Abra no meio da metade posterior do maço
11      Volte para a linha 3
12  Senão
13      Desista
```

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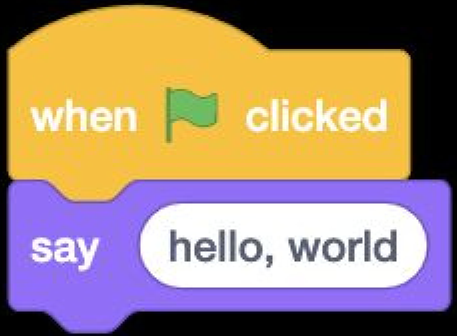
110111100000111000001101100101111100110110001101000010110101101100011010010110111001110101011110000010
110101100111011011100111010100101111011011000110010000101101011011000110100101101110011101010111100000
101101011110000011100000110110001011010011011000110100...

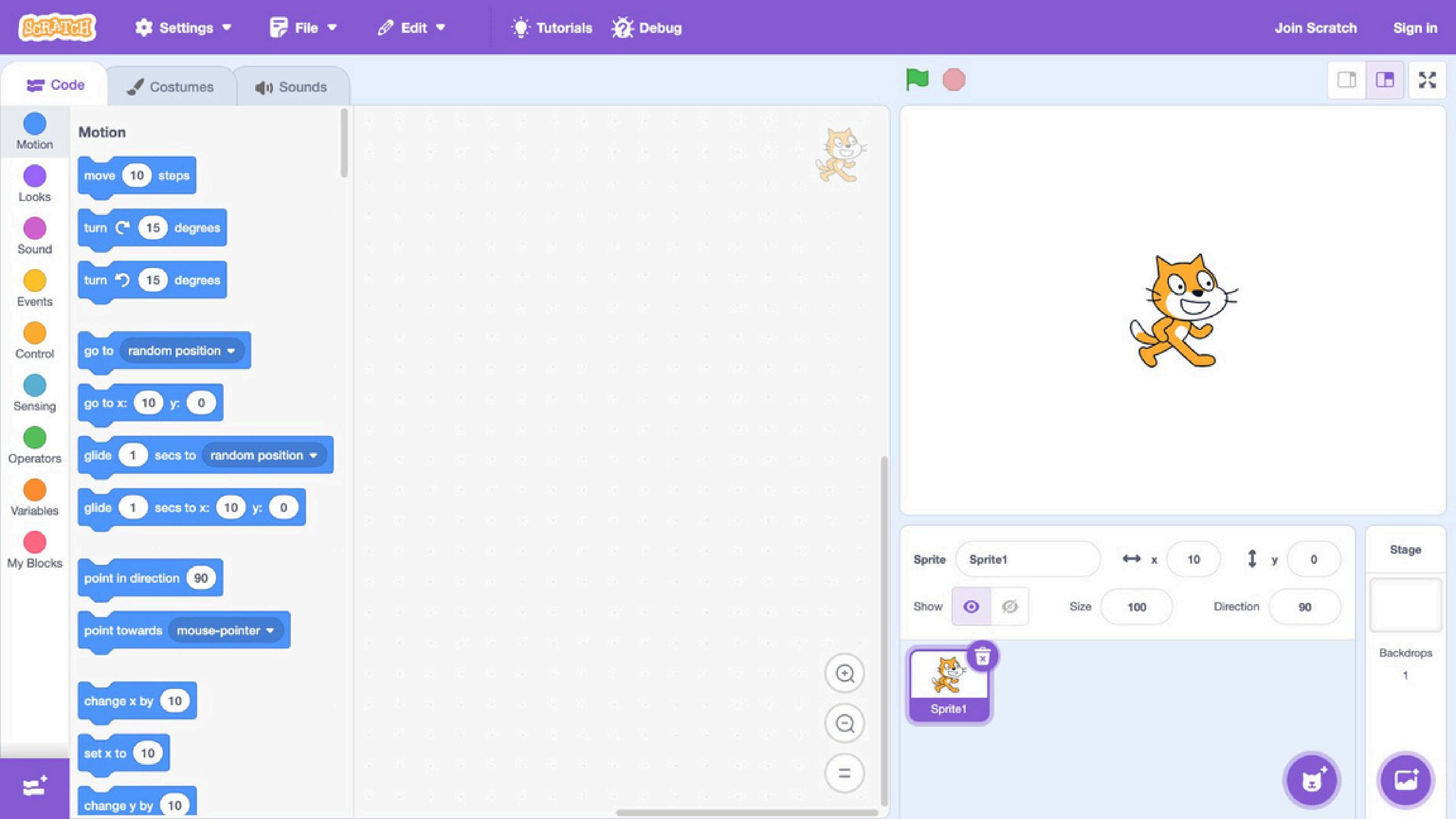
```
#include <stdio.h>

int main(void)
{
    printf("hello, world\n");
}
```

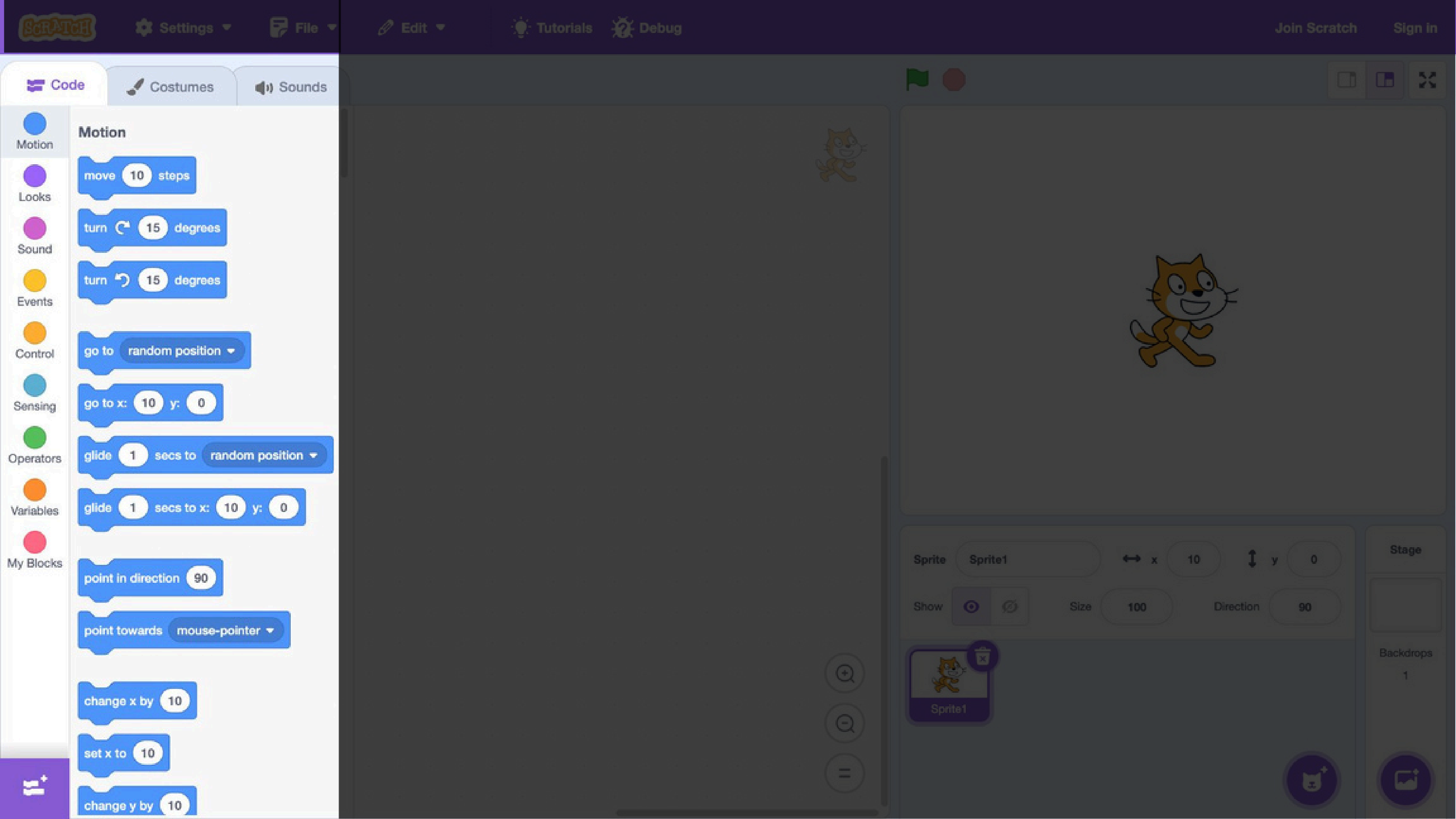

when clicked
say hello, world

Scratch

scratch.mit.edu



Settings
File
Edit
Tutorials
Debug
Join Scratch
Sign in
Code
Costumes
Sounds
Motion
Looks
Sound
Events
Control
Sensing
Operators
Variables
My Blocks
Motion
move 10 steps
turn 15 degrees
turn 15 degrees
go to random position
go to x: 10 y: 0
glide 1 secs to random position
glide 1 secs to x: 10 y: 0
point in direction 90
point towards mouse-pointer
change x by 10
set x to 10
change y by 10
Sprite
Sprite1
x
10
y
0
Show
Size
100
Direction
90
Sprite1
Stage
Backdrops
1


Settings
File
Edit
Tutorials
Debug
Join Scratch
Sign in
Code
Costumes
Sounds
Motion
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Control
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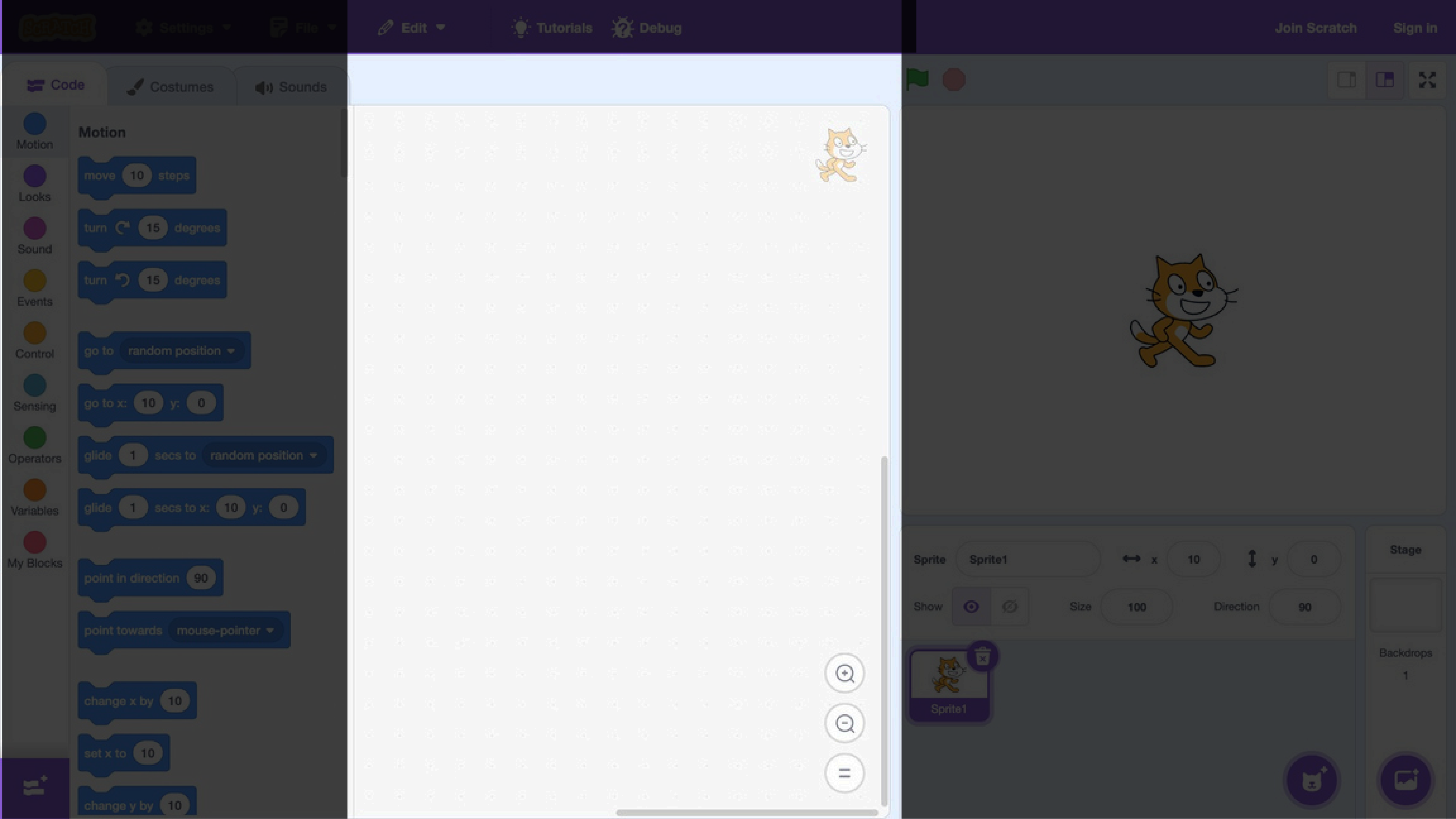
Show
Size
100
Direction
90
Stage
Backdrops
1
Sprite1



Edit
Tutorials
Debug
Join Scratch
Sign in
Code
Costumes
Sounds
Motion
Looks
Sound
Events
Control
Sensing
Operators
Variables
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Direction
90
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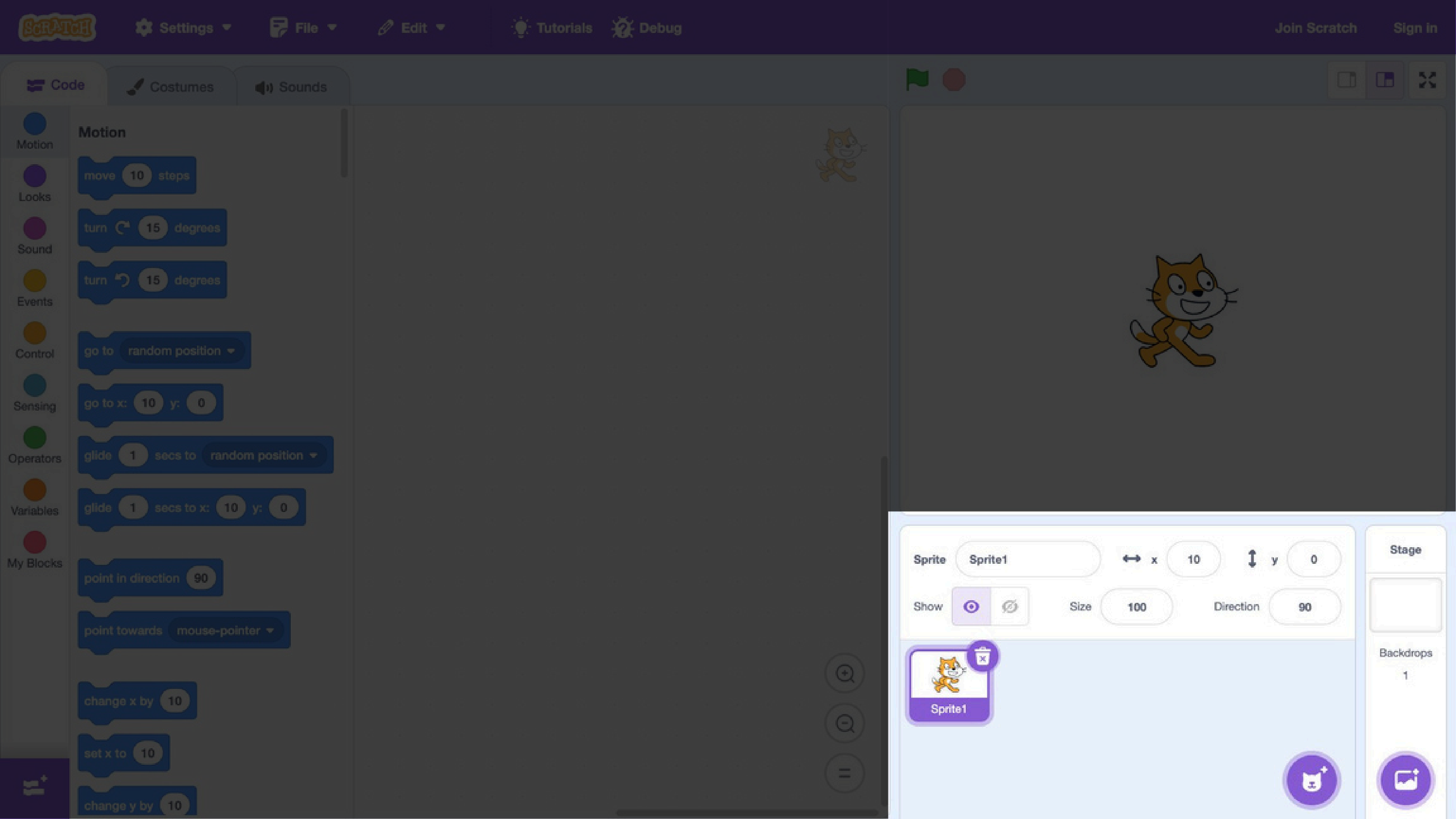
1
Sprite1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File
Edit
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Debug
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Control
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point towards mouse-pointer
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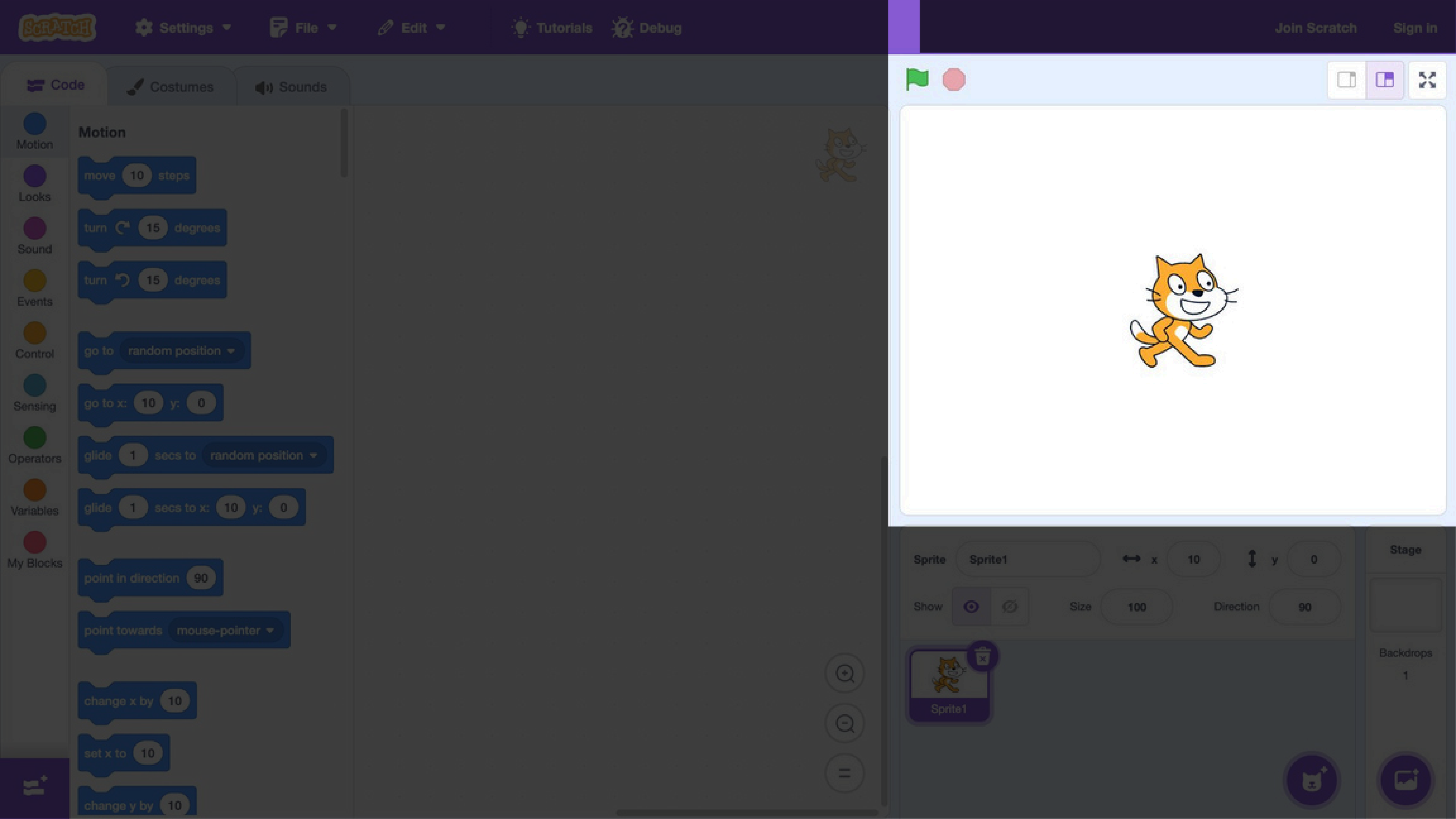
set x to 10
change y by 10
Sprite Sprite1
x 10
y 0
Show
Size 100
Direction 90
Sprite1
Stage
Backdrops
1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File
Edit
Tutorials
Debug
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Costumes
Sounds
Motion
Looks
Sound
Events
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Sensing
Operators
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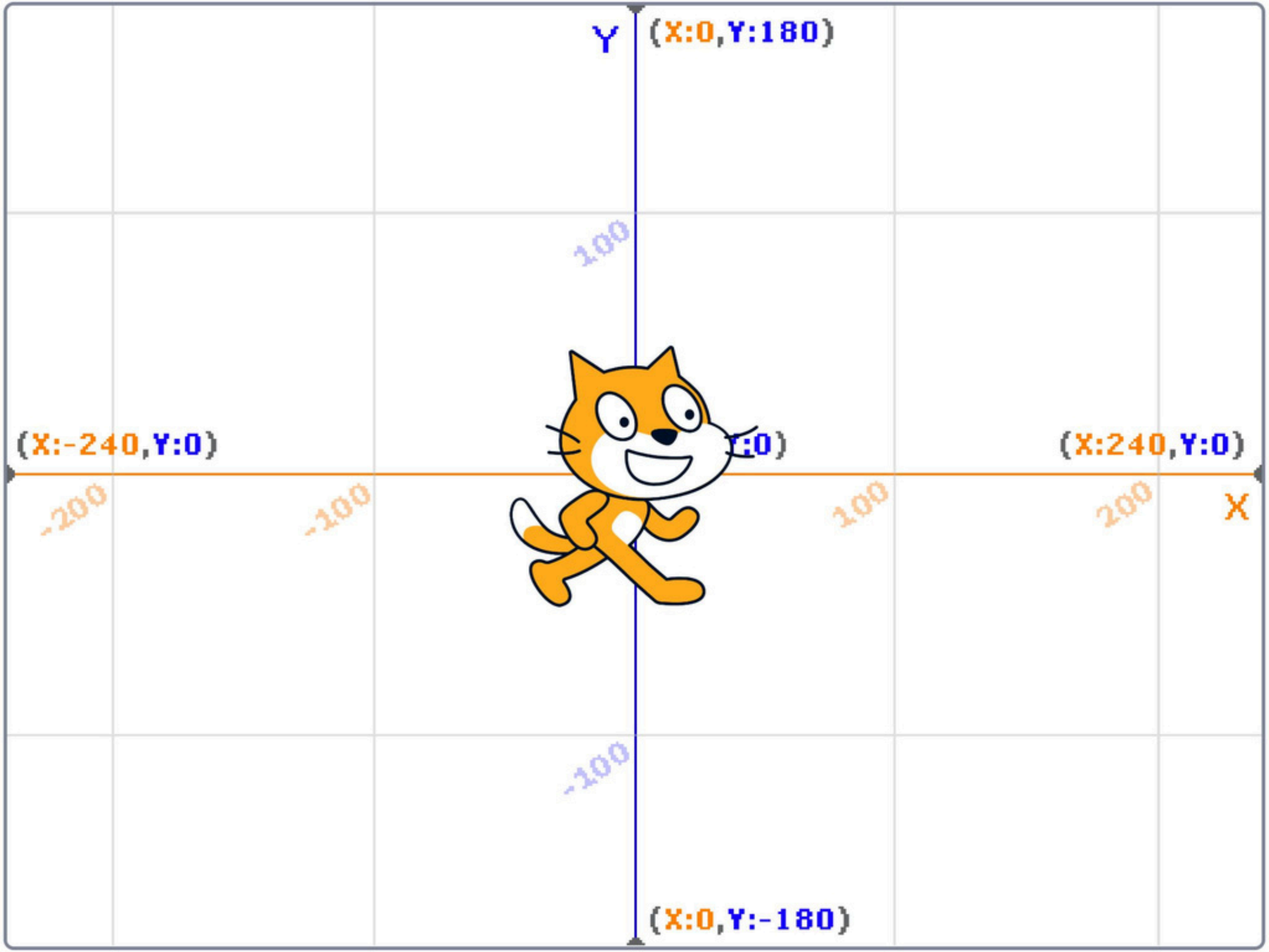


Y
(X:0,Y:180)
100
(X:-240,Y:0)
Y:0)
(X:240,Y:0)
-200
-100
100
200
X
-100
(X:0,Y:-180)

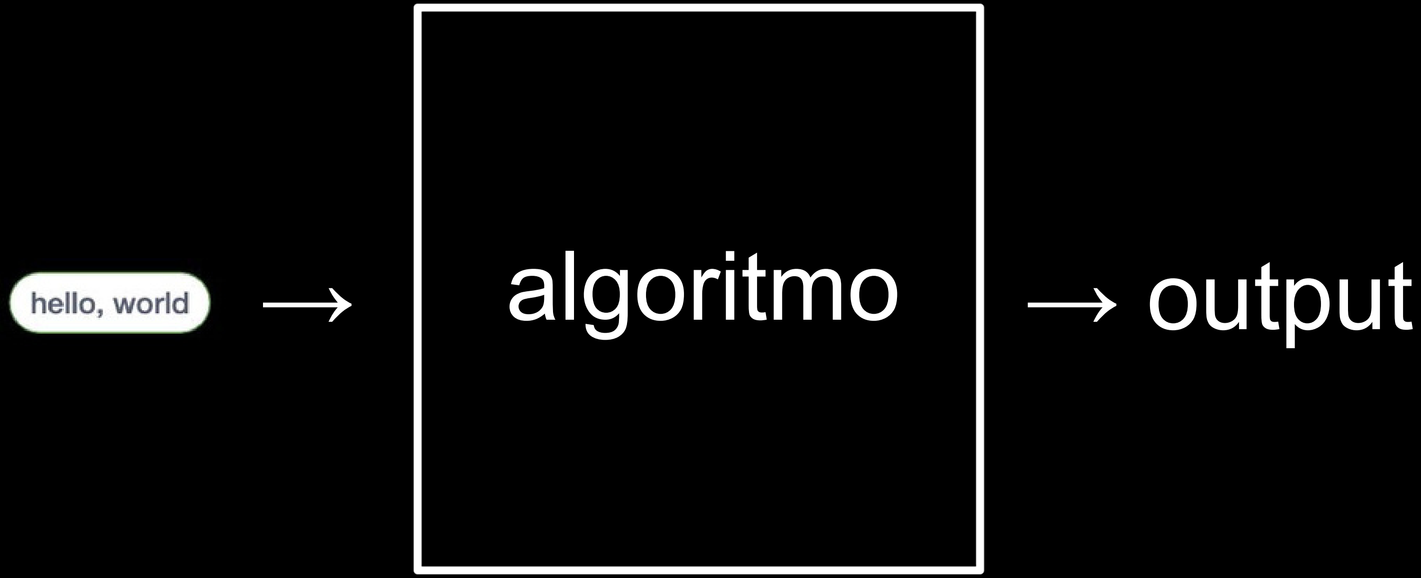
funções


say
hello, world

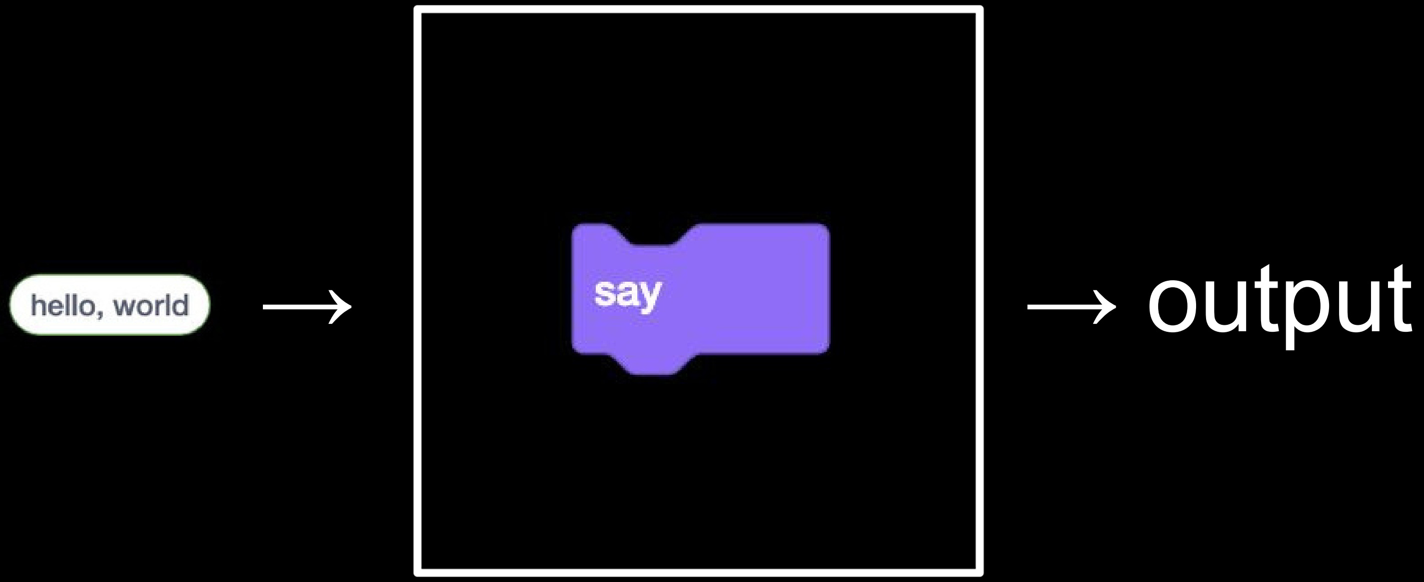
argumentos

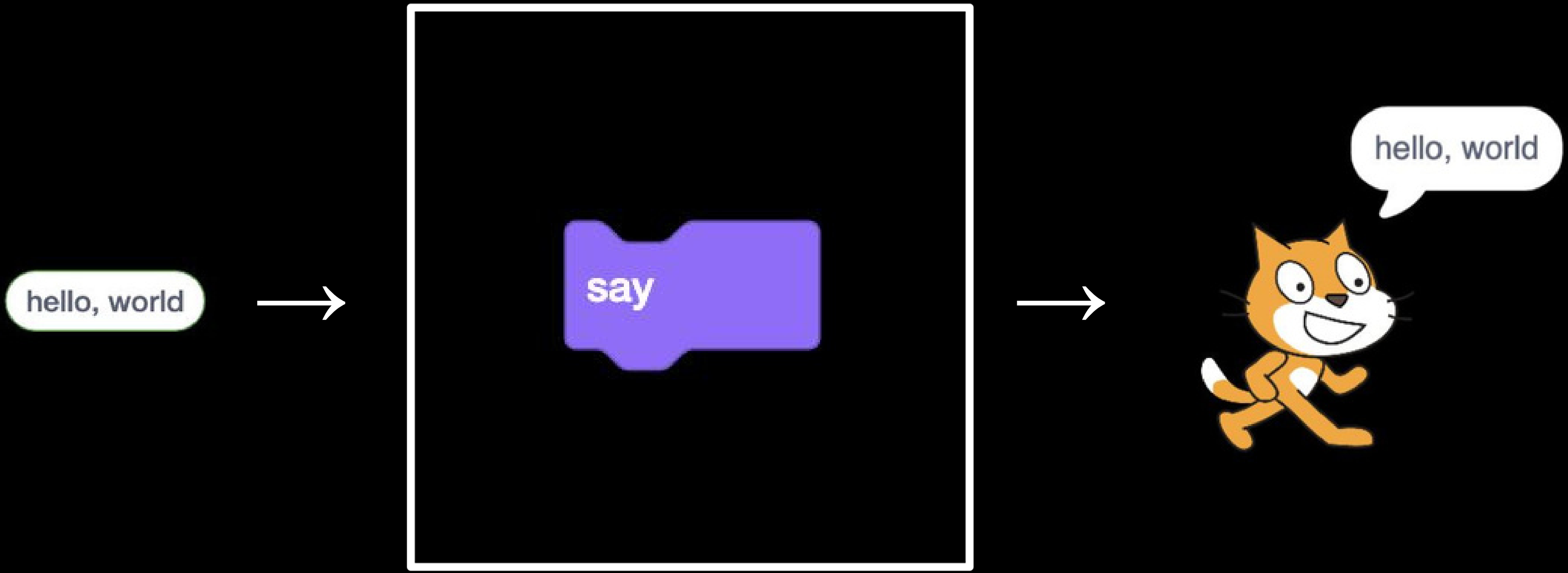
efeitos colaterais

input → algoritmo → output



hello, world
→
algoritmo
→ output


hello, world
say
output

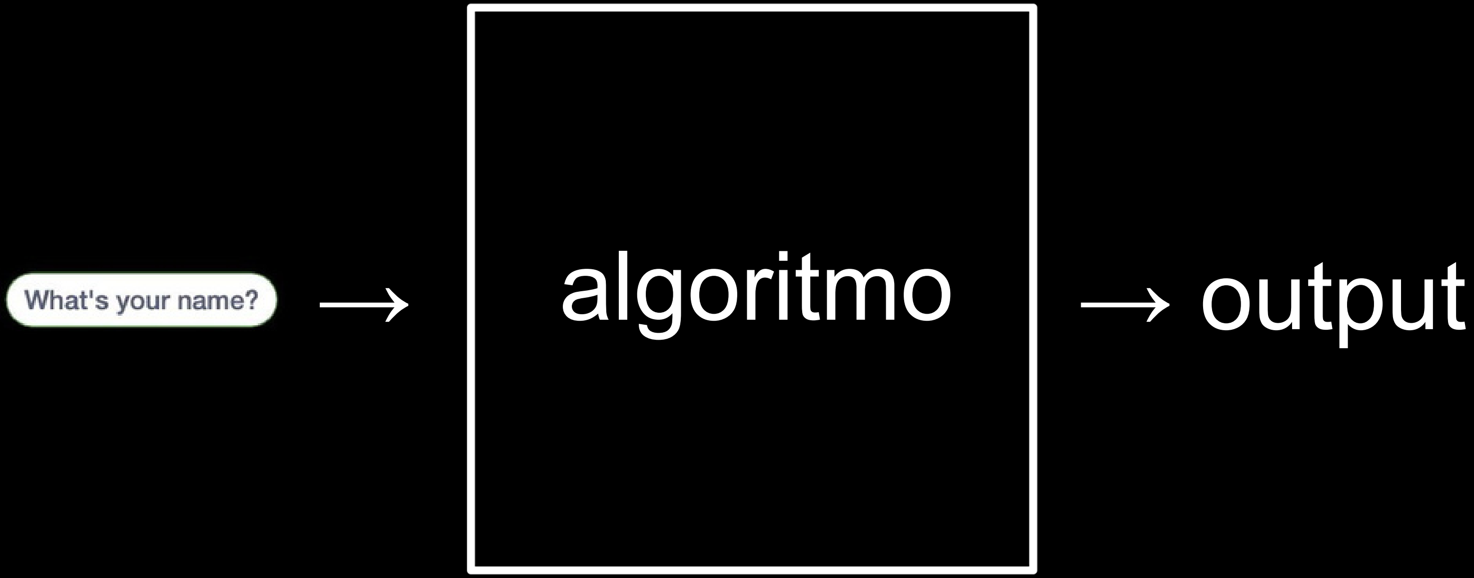

hello, world
say
hello, world

ask What's your name? and wait

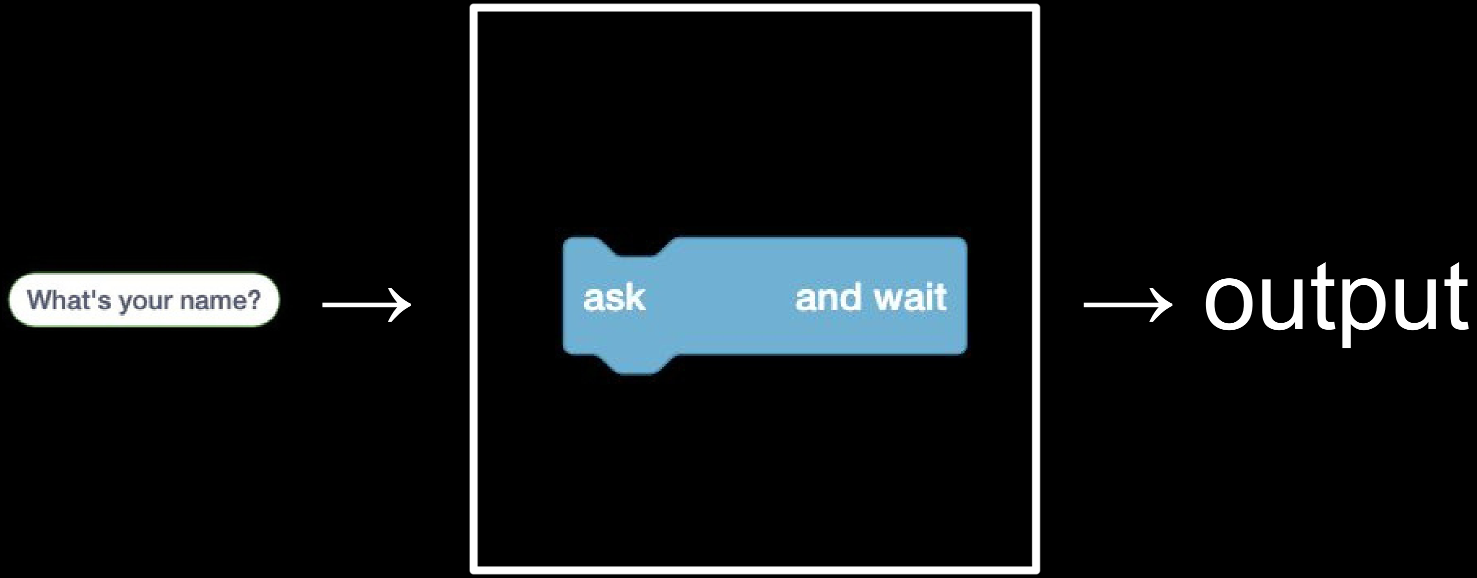
valores de retorno

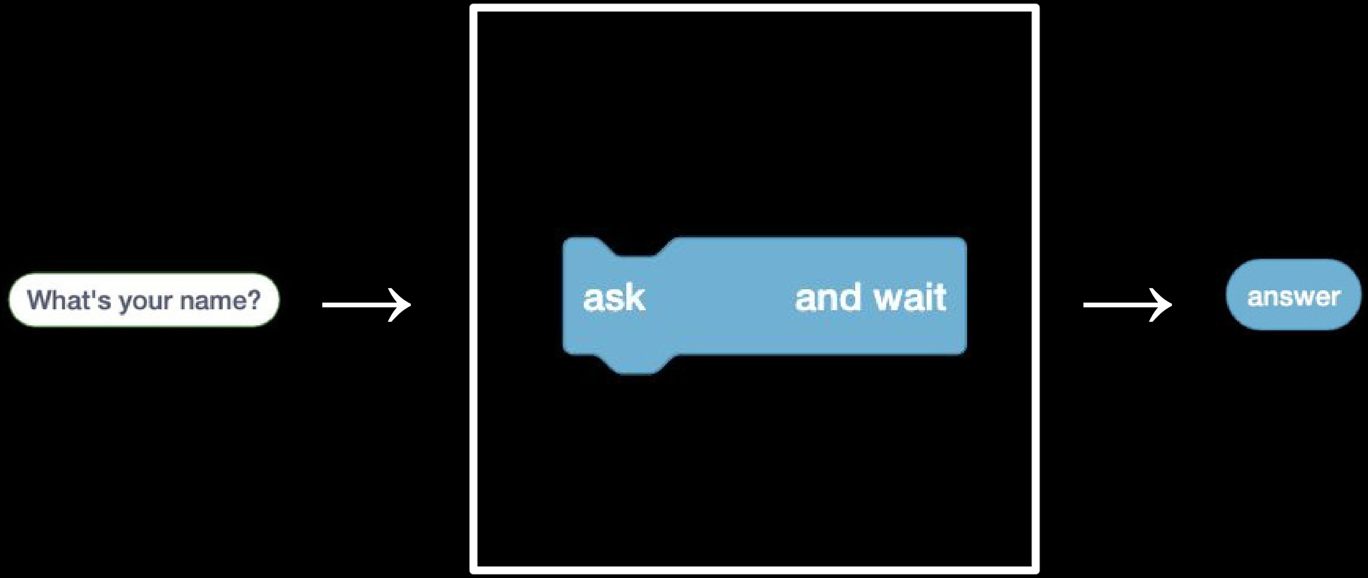
variáveis

input → $\boxed{\text{algoritmo}}$ → output



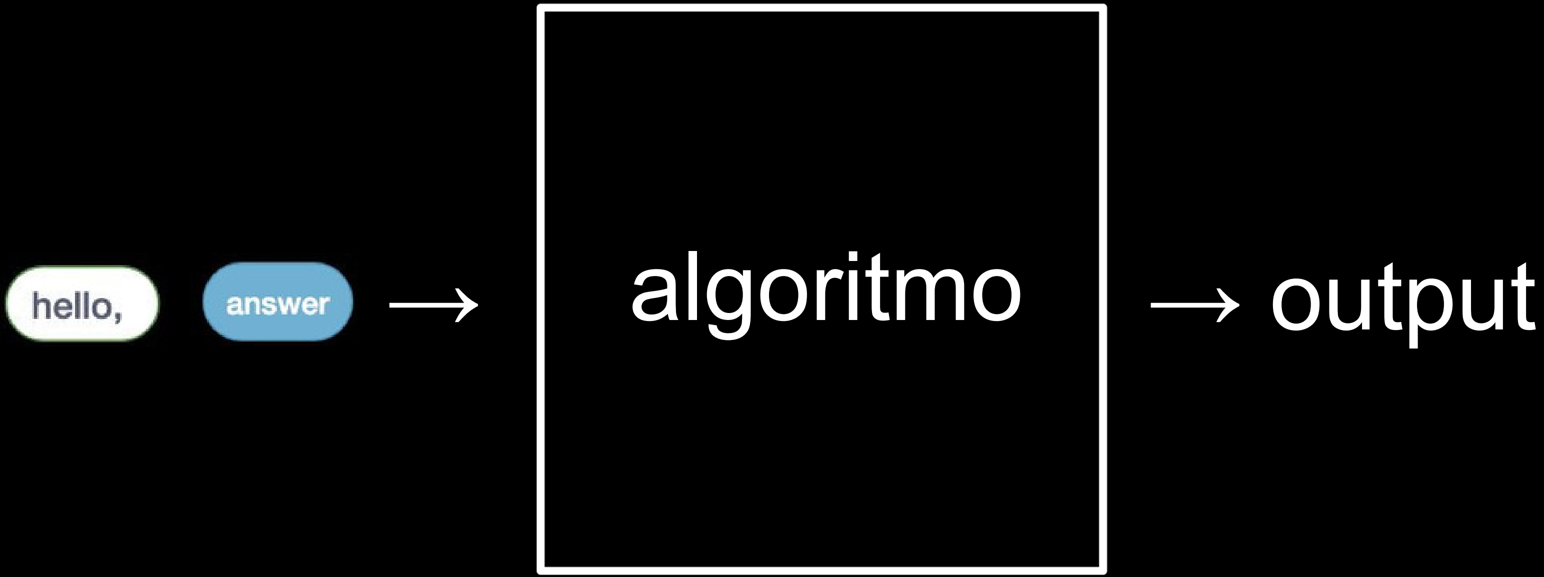
What's your name?
algoritmo
output

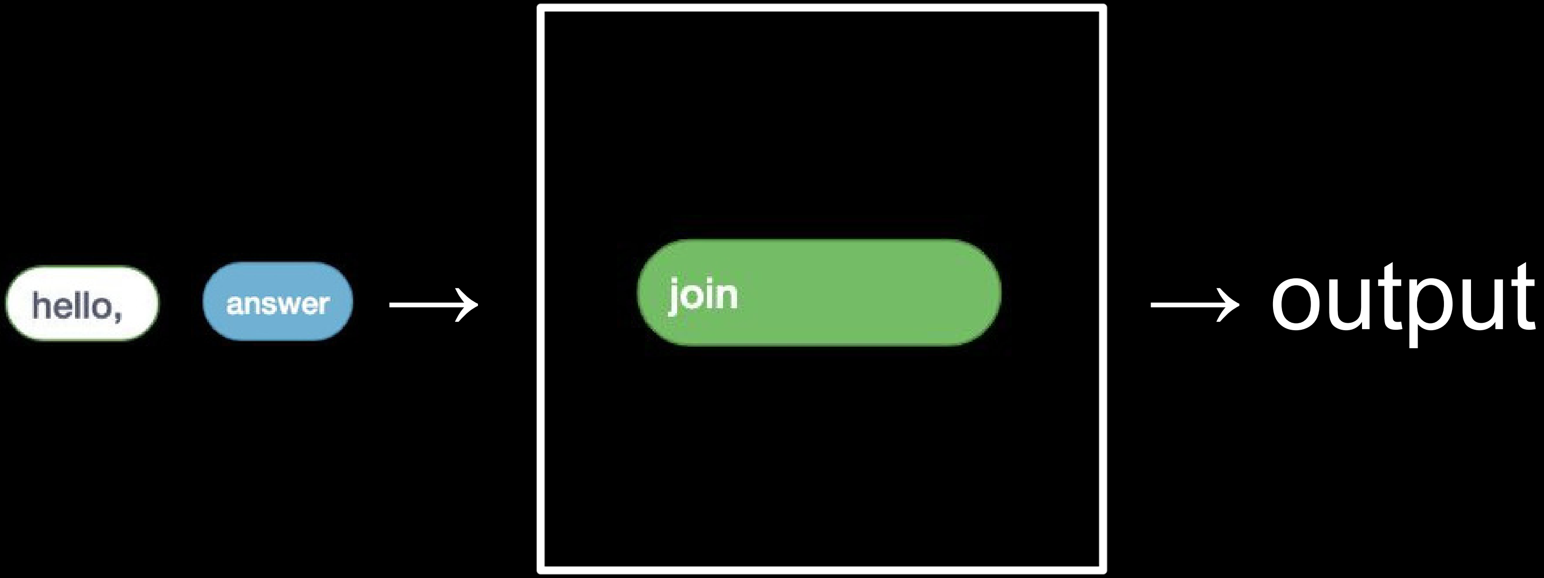

What's your name?
ask
and wait
output

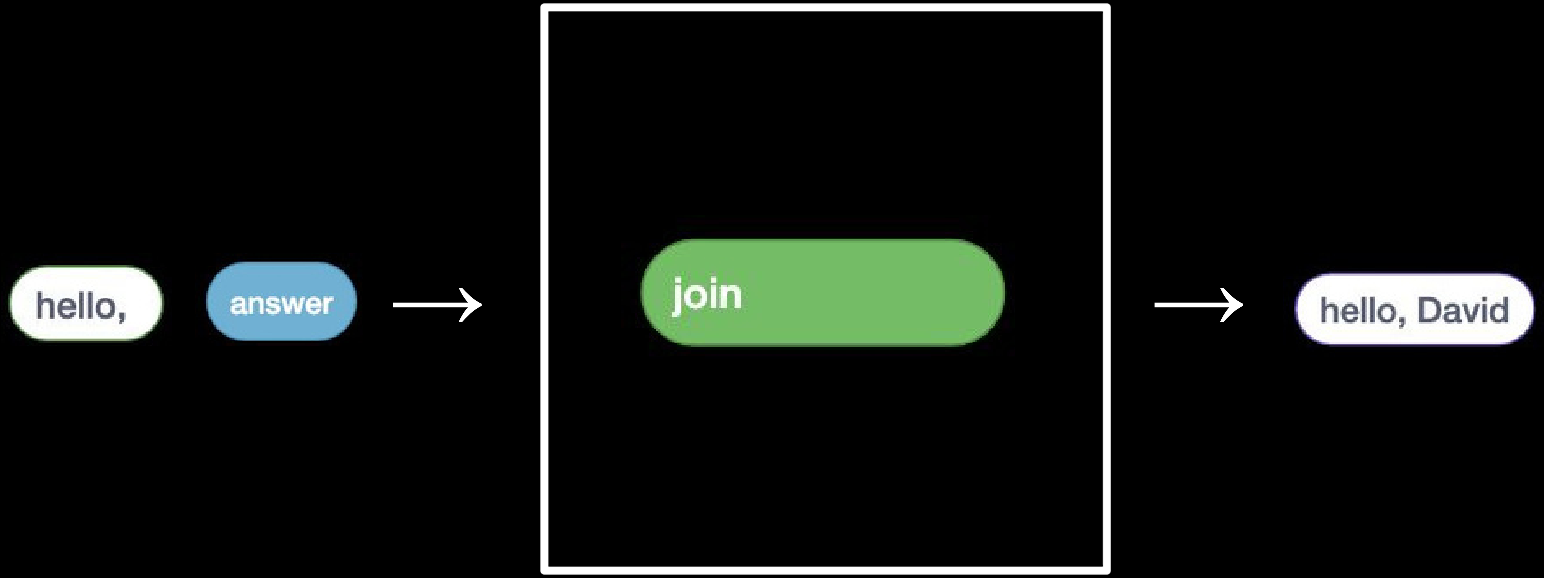

What's your name?
ask and wait
answer


say
join
hello,
answer

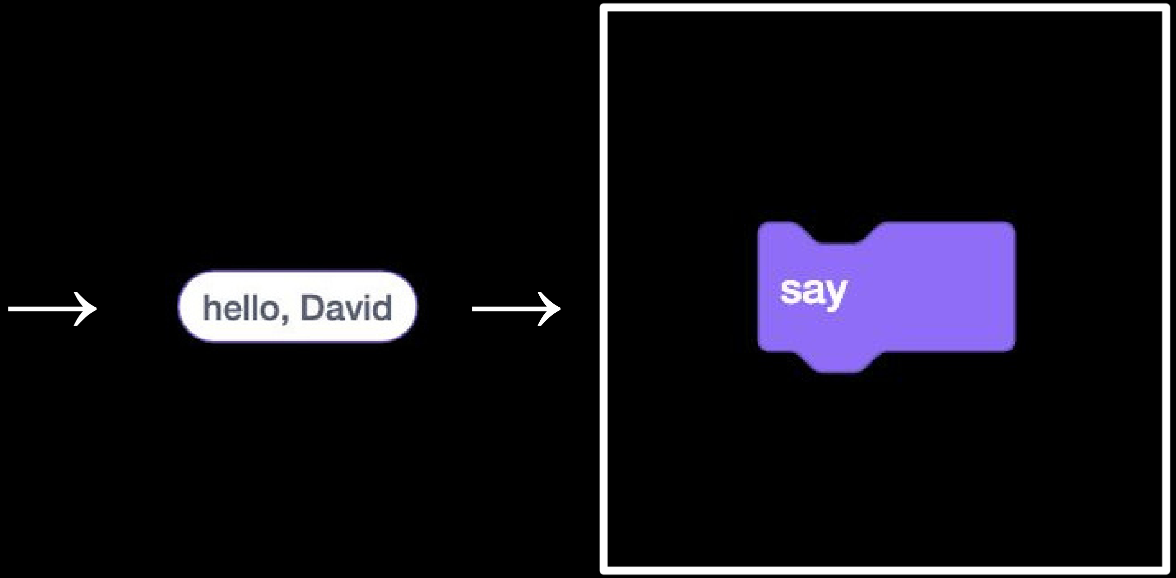
input → $\boxed{\text{algoritmo}}$ → output


hello,
answer
algoritmo
output


hello,
answer
join
output


hello,
answer
join
hello, David

→ hello, David


hello, David
say


hello, David
say
hello, David

laços de repetição

loops

abstração

condicionais

expreções lógicas

expressões booleanas

boolean expressions

Competição